



Python Programming - 2301CS404

Lab - 10

Roll No : 315

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01) WAP to read and display the contents of a text file. (also try to open the file in some other directory)

- in the form of a string
- line by line
- in the form of a list

```
In [16]: ## tell() for cursor reading ,  
# seek(offset , from-where) -->from where : 0 - from begining of the file  
# 1-from current position of the file  
# 2 - from the end of the file  
  
fp = open('abc.txt')  
data = fp.read()  
print(str(data))  
fp.close()
```

```
hihi heloo.....  
it is the 2 nd line..
```

```
In [15]: fp = open('abc.txt')  
data = fp.readlines()  
print(data)  
fp.close()
```

```
['hihi heloo.....\n', 'it is the 2 nd line..']
```

```
In [19]: fp = open('abc.txt')
data = fp.read()
print(list(data))
fp.close()
```

```
['h', 'i', 'i', 'i', ' ', 'h', 'e', 'l', 'o', 'o', '.', '.', '.', '.', '\n',
'i', 't', ' ', 'i', 's', ' ', 't', 'h', 'e', ' ', '2', ' ', 'n', 'd', ' ', 'l', 'i',
'n', 'e', '.', '.']
```

02) WAP to create file named "new.txt" only if it doesn't exist.

```
In [22]: fp = open('new.txt', 'w')
fp.close()
```

03) WAP to read first 5 lines from the text file.

```
In [28]: fp = open('abc.txt')

for i in range(5):
    data = fp.readline()
    print(data, end='')
fp.close()
```

```
hihi heloo.....
it is the 2 nd line..
it is the 3 rd line..
it is the 4 th line..
it is the 5 th line..
```

04) WAP to find the longest word(s) in a file.

```
In [31]: with open("abc.txt", "r") as f:
words = f.read().split()

longest_word = max(words, key=len)
print(longest_word)
f.close()
```

```
heloo.....
```

05) WAP to count the no. of lines, words and characters in a given text file.

```
In [40]: line_count = 0
word_count = 0
char_count = 0

with open("abc.txt", "r") as file:
    for line in file:
        line_count += 1
        char_count += len(line)
        words = line.split()
```

```

        word_count += len(words)

print("Number of lines:", line_count)
print("Number of words:", word_count)
print("Number of characters:", char_count)
f.close()

```

Number of lines: 5
 Number of words: 26
 Number of characters: 103

06) WAP to copy the content of a file to the another file.

```

In [42]: fp = open('abc.txt')
        data = fp.read()
        fp1 = open('new.txt', 'w')
        fp1.write(data)
        fp.close()
        fp1.close()
        print('file copy sucessfully')

```

file copy sucessfully

07) WAP to find the size of the text file.

```

In [4]: fp = open('abc.txt')
        fp.read()
        print(fp.tell())
        fp.close()

```

15

08) WAP to create an UDF named frequency to count occurances of the specific word in a given text file.

```

In [50]: word = input('enter word')
        def findOcuurnce(word):
            fp = open('abc.txt')
            print(fp.read().lower().count(word))
            fp.close()

        findOcuurnce(word)

```

4

09) WAP to get the score of five subjects from the user, store them in a file. Fetch those marks and find the highest score.

```

In [51]: with open("marks.txt", "w") as f:
        for i in range(5):
            mark = int(input(f"Enter marks for subject {i+1}: "))
            f.write(str(mark) + "\n")

```

```

with open("marks.txt", "r") as f:
    marks = f.read().split()

marks = list(map(int, marks))

highest = max(marks)

print("Highest score is:", highest)

```

Highest score is: 20

10) WAP to write first 100 prime numbers to a file named primenumbers.txt

(Note: each number should be in new line)

```

In [53]: def is_prime(n):
        if n < 2:
            return False
        for i in range(2, int(n ** 0.5) + 1):
            if n % i == 0:
                return False
        return True

with open("primenumbers.txt", "w") as f:
    count = 0
    num = 2

    while count < 100:
        if is_prime(num):
            f.write(str(num) + "\n")
            count += 1
        num += 1
    print('wriiten sucessfully')

```

wriiten sucessfully

11) WAP to merge two files and write it in a new file.

```

In [55]: with open("abc.txt", "r") as f1, open("new.txt", "r") as f2:
        data1 = f1.read()
        data2 = f2.read()

with open("merged.txt", "w") as f:
    f.write(data1)
    f.write(data2)
    print('merge sucessfully')

```

merge sucessfully

12) WAP to replace word1 by word2 of a text file. Write the updated data to new file.

```
In [57]: with open("abc.txt", "r") as f:
          data = f.read()

          data = data.replace("its", "here it's")

          with open("new.txt", "w") as f:
              f.write(data)
```

13) Demonstrate tell() and seek() for all the cases(seek from beginning-end-current position) taking a suitable example of your choice.

```
In [60]: f = open("abc.txt", "rb")

          print("Initial position:", f.tell())

          print(f.read(5))
          print("After reading 5 bytes:", f.tell())

          f.seek(0, 0)
          print("\nSeek from beginning")
          print("Position:", f.tell())
          print(f.read(5))

          f.seek(3, 1)
          print("\nSeek from current position")
          print("Position:", f.tell())
          print(f.read(5))

          f.seek(-10, 2)
          print("\nSeek from end")
          print("Position:", f.tell())
          print(f.read())

          f.close()
```

```
Initial position: 0  
b'hiii '  
After reading 5 bytes: 5
```

```
Seek from beginning  
Position: 0  
b'hiii '
```

```
Seek from current position  
Position: 8  
b'oo...'
```

```
Seek from end  
Position: 93  
b' th line..'
```

In []: